

30-10-25

convolution layer consist of 4 why we use it.

DL - Keras with TensorFlow

Sequential

Functional

Model subclassing

MNIST

It will auto detect
which kernel
to use
Kernel $\rightarrow$ FILTERS

model = Sequential(
 7 32 neuron

$\begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \times 32$

model.add(Conv2D(32, (3,3), activation='relu',
 input_shape=(28,28,1)

model.add(MaxPooling2D(2,2))

(model.add(Conv2D(64, (3,3), activation='relu')
 model.add(MaxPooling(2,2))

model.add(Flatten()),

model.add(Dense(64, activation='relu')

model.add(Dense(10, activation='Softmax')

Kernel is same as filter.

MNIST

28 $\begin{bmatrix} 0 \\ 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}$ - - - -

pixels from 0 to 255 $\rightarrow$ scaling 0 to

$\begin{bmatrix} 0 \\ 1 \\ 2 \\ 3 \end{bmatrix} \rightarrow \begin{bmatrix} 255 & 255 & \dots & \dots \\ 255 & 0 & \dots & \dots \\ 255 & \dots & \dots & \dots \end{bmatrix} \times 255 \begin{bmatrix} 1 & 0.9 \\ 0 & \dots \end{bmatrix}$